Multilayer sticker-type anti-reflective film for high-efficiency optoelectronic devices

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고효율 광전자소자를 위한 다층 스티커형 반사방지 필름

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Abstract

Conventional anti-reflective films are widely used in high-performance optoelectronic devices. Despite significant advances in internal material optimization, efficient light coupling at external interfaces remains a critical limitation. To address this challenge, we present a sticker-type multilayer anti-reflective (SMAR) film that integrates optical properties with surface adhesion. The film consists of a bilayer architecture of a nanostructured perfluoropolyether (PFPE) layer and adhesive polydimethylsiloxane (PDMS) layer. This architecture provides a gradual refractive index transition from air to the substrate. A simple ethanol-assisted separation strategy was introduced to fabricate free-standing multilayer films without damage. The SMAR film improves the transmittance of polyethylene naphthalate (PEN) substrates to 94.81% and reduces reflectance to 5.05%. When applied to flexible perovskite solar cells, the film increases the short-circuit current density and improves the power conversion efficiency from 21.31% to 22.61%. This work demonstrates a practical and reusable optical interface for advanced flexible optoelectronic devices.

Keywords: ***Anti-reflective, Sticker-type, Gradual refractive index, Surface adhesion, Optoelectronic devices***

1. Introduction

Efficient light management is a key requirement for high-performance optoelectronic devices such as photovoltaics, light-emitting diodes, and photodetectors [1-3]. Although significant progress has been achieved through the optimization of internal materials and device structures, optical loss remains a limitation at external interfaces. Reflection is caused by refractive index mismatch between air and substrates. This reduces the amount of light transmitted into active layers and leads to the degradation of device performance.

Anti-reflective (AR) films offer a simple and effective strategy to compensate for this loss [4-5]. However, conventional AR coatings require direct fabrication on device substrates, which can induce process mechanical damage, or limited applicability to flexible optoelectronics. Recent sticker-type AR films provide a non-destructive alternative, yet most single-layer designs suffer from a trade-off between optical performance and adhesion.

In this paper, we present a sticker-type multilayer AR (SMAR) film designed to improve optical performance and surface adhesion for optoelectronic devices. As shown in Fig. 1, the SMAR film employs multilayer architecture consisting of a nanostructured layer that provides a gradual refractive index profile and an adhesive layer.

2. Fabrication and optical performance of SMAR film

2.1 Theory and design

The SMAR film was theoretically designed to optimize both optical performance and mechanical stability for flexible optoelectronic devices. A multilayer refractive index profile was calculated using a nanostructured perfluoropolyether (PFPE) layer and a polydimethylsiloxane (PDMS) adhesive layer between air and substrate. The effective refractive index of the nanostructured layer was designed using a volume-fraction-based effective medium approach and confirmed a continuous index gradient across the optical path. Optical performance was evaluated using Fresnel equations and Rouard's method [6]. These equations predicted a transmittance enhancement of approximately 6% for PEN substrates. These theoretical results guided the material selection and structural parameters of the SMAR film.

2.2 Fabrication process

The SMAR film was fabricated through a soft lithography based multilayer assembly process in Fig. 2. A nanostructured PFPE layer was first prepared using a silicon master mold and UV curing with sacrificial polyethylene terephthalate (PET) film. A thin PDMS adhesive layer was subsequently spin-coated onto the PET film and the PFPE layer was placed onto the uncured PDMS to form the multilayer structure. After thermal curing, an ethanol-assisted separation was employed to enable clean release of the free-standing SMAR film without damaging the PFPE-PDMS interface.

Ethanol selectively infiltrated the PDMS-PET interface by reducing adhesion at the sacrificial interface while preserving strong interlayer bonding.

2.3 Optical performances

The optical performance of the SMAR film was evaluated by measuring transmittance and reflectance in the visible wavelength range (400-800 nm) on both glass and PEN substrates (Fig. 3a-b). Compared to bare substrates and single-layer PFPE films, the SMAR film exhibited improvement of optical performance due to the gradual refractive index transition and enhanced adhesion to substrate by the PDMS adhesive layer. On PEN substrates, the transmittance increased from 89.05% to 94.81%, while reflectance decreased to 5.05%. The optical enhancement was maintained across various SMAR film thicknesses, which was governed primarily by the nanostructured surface rather than bulk thickness. These results validate the effectiveness of the multilayer optical design.

2.4 Mechanical durability and applicability

The mechanical durability and practical applicability of the SMAR film were evaluated through repeated attachment/ detachment tests and photovoltaic demonstrations as shown in Fig. 3c-d. The film maintained its optical performance after 10,000 cycles of reattachment and repeated bending at a radius of 2 mm. When applied to flexible perovskite solar cells, the enhanced light coupling resulted in a clear improvement in device performance with short-circuit current density increasing from 23.81 to 25.26 mA/cm^2 and power conversion efficiency improving from 21.31% to 22.61%, while open circuit voltage and fill factor remained unchanged. These results demonstrate that the SMAR film functions as a practical and reusable optical interface for improving device performance while maintaining mechanical reliability.

Acknowledgments

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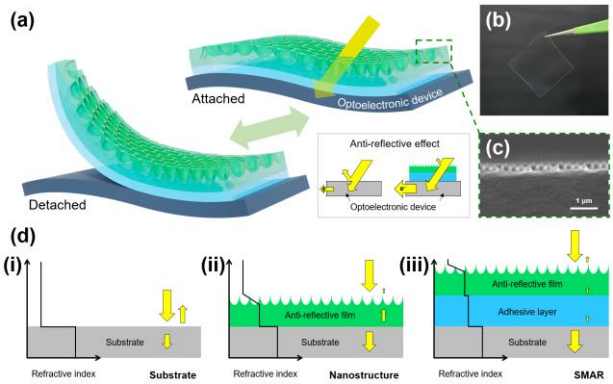

Fig. 1. Damage-free and repeatable adhesion of the SMAR film on an optoelectronic device.

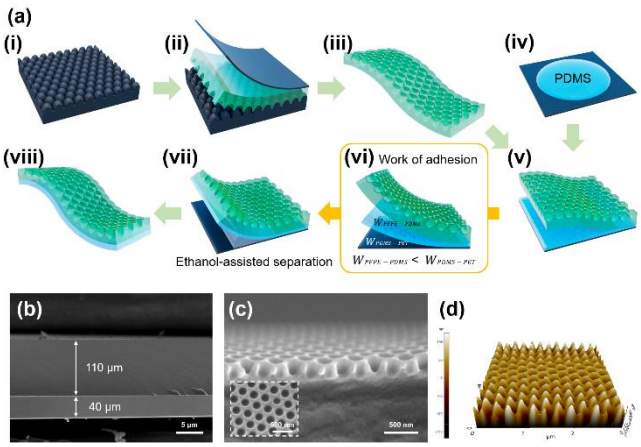

Fig. 2. Illustrated schematics of fabrication methods of the SMAR film based on the PFPE and PDMS film.

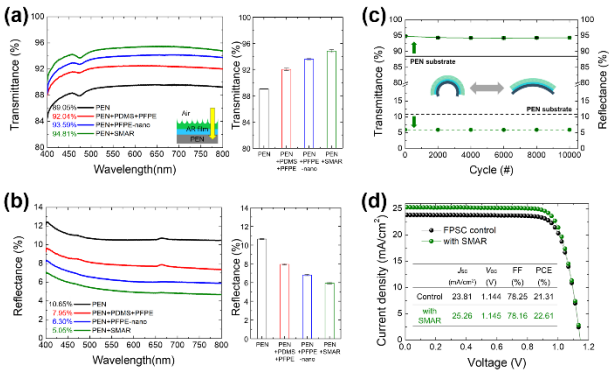

Fig. 3. Optical performances and device applicability of SMAR film.